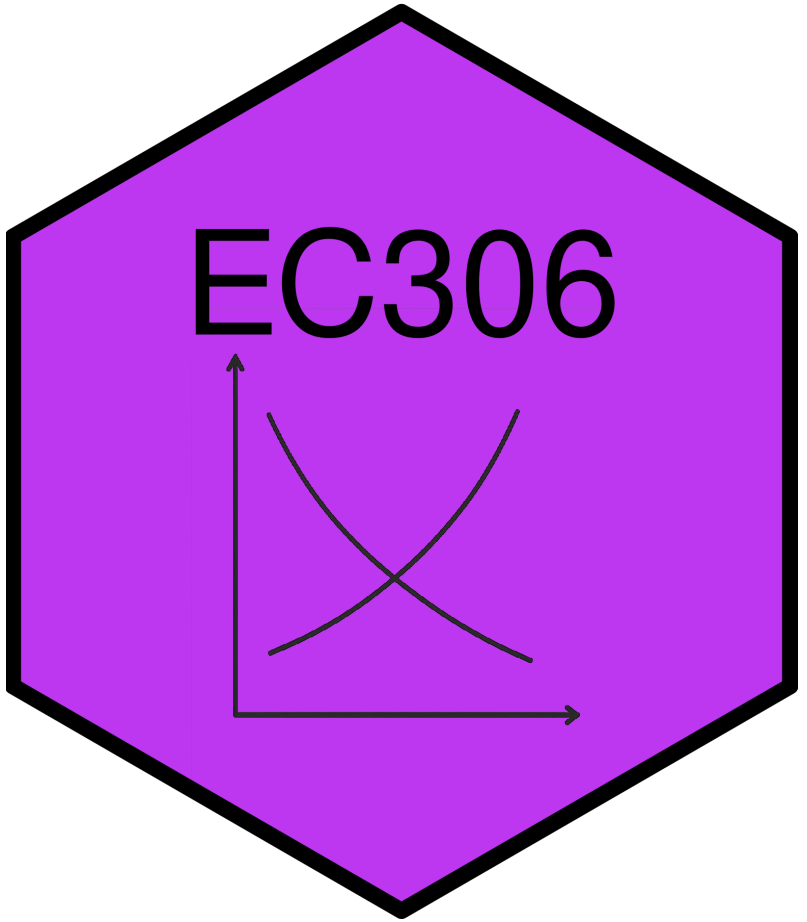
Labour Supply and Income Support Programs

EC306- Wages and Employment

Justin Smith

Wilfrid Laurier University

Fall 2026



Goals of This Section

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In this section we will:

- Describe the main income support programs in Canada
- Use the labour supply model to evaluate a policy, step by step
- Analyze the work incentives of demogrants, social assistance, and negative income taxes
- Analyze wage subsidies and refundable tax credits like the Canada Workers Benefit
- Examine how economists measure these effects, including with experiments

Income Support Programs in Canada

Income Support and the Social Safety Net

Key Definitions

Income support programs: cash transfers meant to guarantee a minimum standard of living to people with low income, the core of Canada's **social safety net**

Means-tested: eligibility depends on income or wealth

- Some programs target low income that is permanent, others low income that is temporary
- Social insurance is a separate group: EI, CPP, disability
  - It pays out based on past contributions, not current need, and we cover it in Chapter 5
- Both are income maintenance programs: they support incomes, and they change how much people choose to work
- That is the tension in this chapter: support versus work incentives

Government Transfers in Canada, 2022

In 2022, 95% of Canadian families received at least one government transfer

Benefit	Percent of families	Average amount
All government transfers	**95.1**	**$13,969**
Federal child benefit	21.5	$6,166
Provincial child benefit	12.0	$2,349
GST credit	51.8	$692
Social assistance	8.2	$11,270
Canada Workers Benefit	11.5	$1,347
Federal refundable tax credits	52.6	$677
Old Age Security	26.4	$10,122
Guaranteed Income Supplement	10.2	$6,900

Source: Canadian Income Survey (2022). Census families; average amount among families receiving the benefit.

Issues in Program Design

Universal vs. Targeted Programs

- Universal: easy to implement but expensive
- Targeted: benefits only those in need but costly to administer
- Canada swapped the universal UCCB for the income-tested CCB in 2016

Work Disincentives

- Some programs reduce incentive to work
- Example: welfare benefits can encourage labour force dropout for those with strong preference for leisure

Permanent vs. Transitory Income Loss

- Different situations require different program designs
- Temporary layoff → short-term support
- Long-term disability → longer support + reintegration costs

Think About It

If 95% of Canadian families receive some form of government transfer, why is program design so important? What happens if a program is too generous or too stingy?

Demogrants and Social Assistance Programs

Using the Model to Evaluate a Policy

- Same tool as Chapter 3: comparative statics, comparing choices before and after a change
- Policy rarely changes what people like; it changes what they can have
  - So the action is in the potential income constraint, not the indifference curves
- Every program in this chapter gets the same three steps:
  1. Draw the constraint without the policy
  2. Draw the constraint with the policy
  3. Given preferences, compare the chosen bundles

Key Definitions

Partial equilibrium effects: how the people a program reaches respond, holding wages and everything else fixed

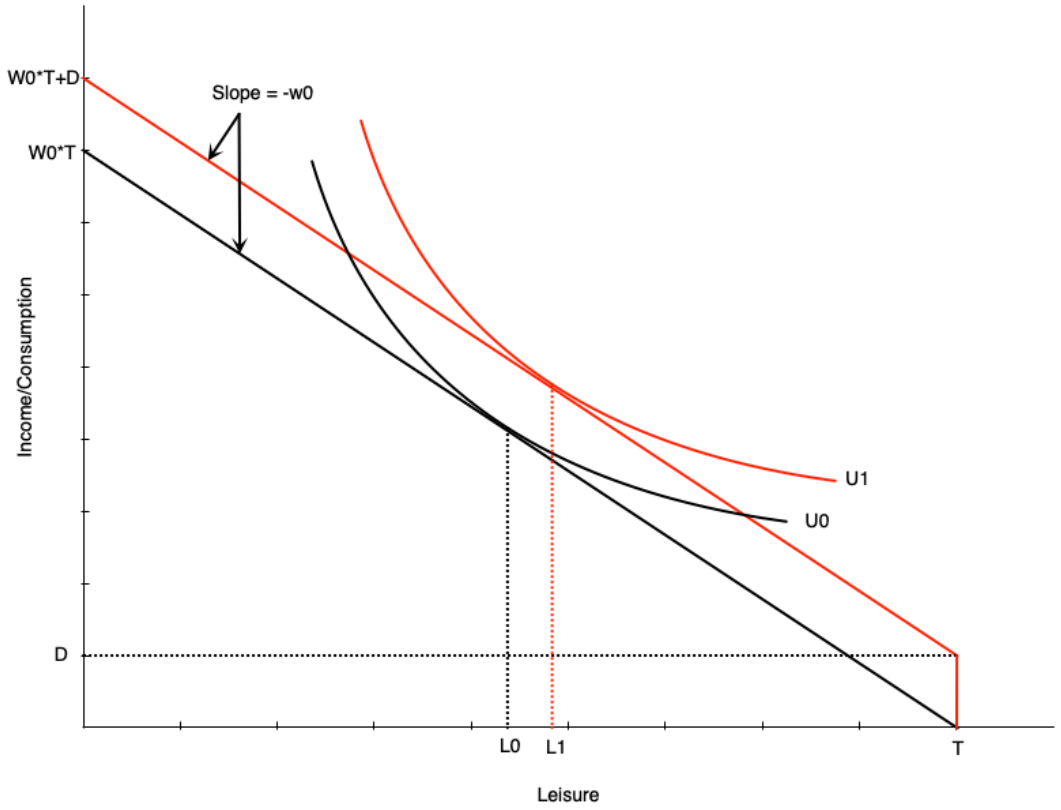
General equilibrium effects: the knock-on effects once the program works its way through the whole economy

Four Standard Income Support Programs

We will examine four policy approaches:

1. Demogrants - Fixed payment to a demographic group, such as everyone over 65, not means-tested
2. Social assistance (welfare) - Tops up income to a threshold for those with little else, clawed back dollar-for-dollar with earnings
3. Negative Income Tax (NIT) - A guarantee clawed back at a rate < 100% as earnings rise
4. Wage Subsidies - Addition to the wage rate

Demogrant: Theory



How it works:

- Unconditional lump sum transfer G
- Shifts budget line up parallel
- Wage unchanged → pure income effect

Labour supply impact:

- If leisure is normal → optimal leisure rises
- Reduces hours worked
- Raises reservation wage → some may quit

⚠ Work Incentive Effect

Unambiguously negative

Demogrants in Canada

Old Age Security (OAS)

- Monthly benefit for people 65 and over who meet residency rules
- Not a pure demogrant: it is taxed back from high-income seniors
- Together with the Guaranteed Income Supplement, more than $76 billion in 2024

Universal Child Care Benefit (UCCB)

- Paid to families with children from 2006 to 2016, whatever their income
  - Started at $100 a month per child under 6
  - From 2015: $160 a month under 6, $60 a month aged 6–17
- Taxable, so lower-income families kept more of it: not a pure demogrant either
- Replaced in 2016 by the income-tested Canada Child Benefit (CCB)

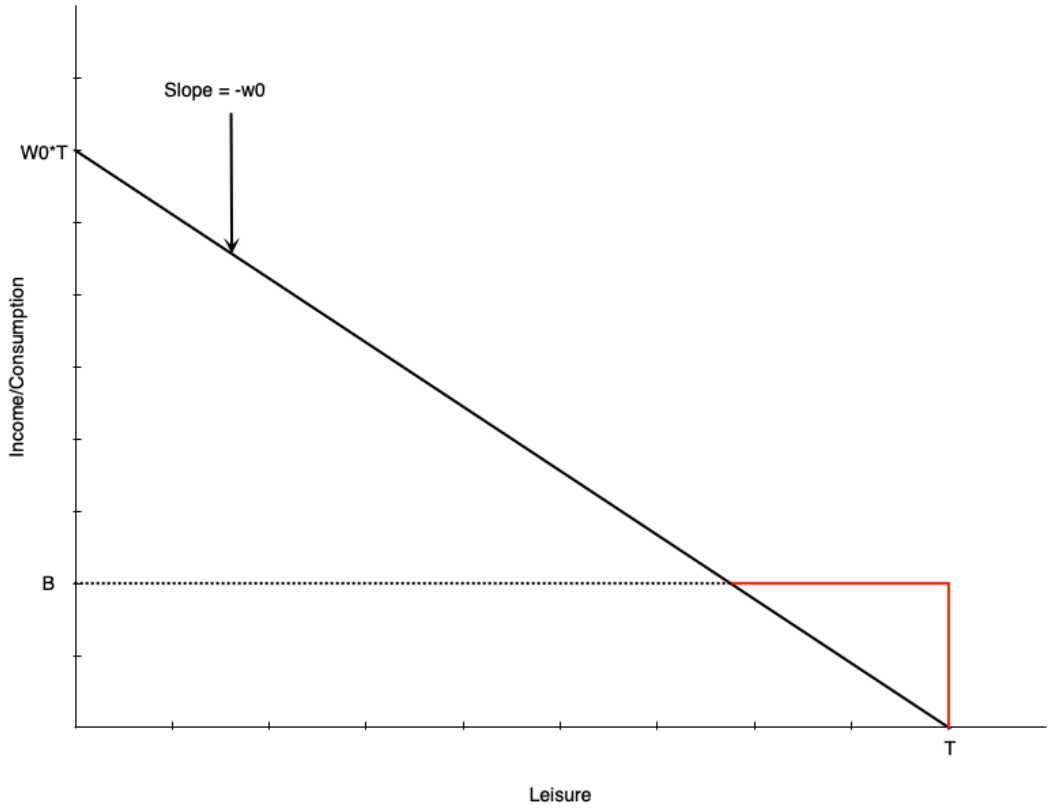
Social Assistance: Overview

Key Definition

Social assistance (income assistance, or **welfare**): means-tested provincial support for people with little or no other income

- Perfect targeting: top income up to a threshold z: grant $G = z - (wH + Y_N)$
- Every dollar earned costs a dollar of benefit: a 100% clawback rate
- Most recipients have a disability or another barrier to full-time work
- Benefits are modest: Ontario pays a single employable adult at most $733 a month (2025), about 10 hours a week at minimum wage
- A lone parent with one child got 57–86% of the poverty line, depending on province (2023)
- Source: Laidley & Tabbara (2024)

Social Assistance: Budget Constraint



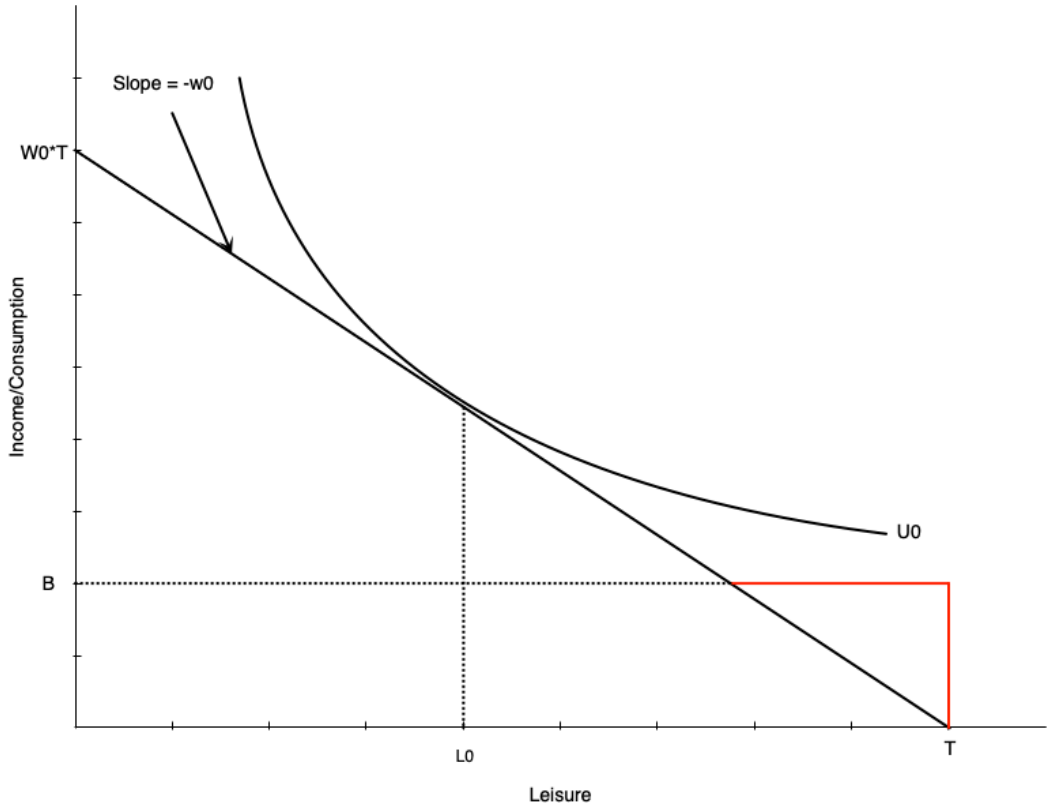
Budget line structure:

- Black line: original budget (leisure at L_0)
- Red then black: with social assistance up to the threshold z

Key features:

- Grant $G = z$ at zero hours of work
- Reduced \$1 for each \$1 earned
  - Equivalent to 100% tax rate on earnings
- Income stays at z until earnings reach z
- After that, earn wage w_0 per hour worked

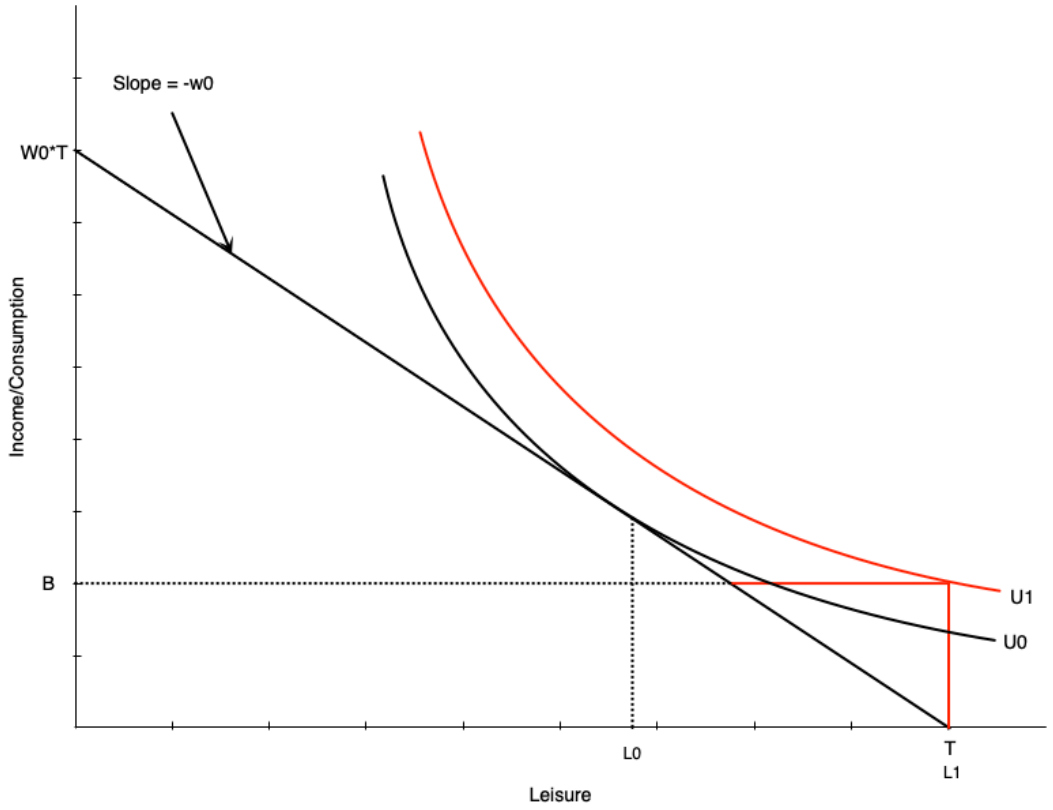
Social Assistance: No Effect Case



For some workers, welfare has no effect:

- Person working H_0 hours before benefit
- Still maximizes utility at H_0 hours after
- Generally occurs with people already working many hours

Social Assistance: Work Disincentive Case



Others face strong work disincentive:

- Person chooses zero hours after welfare benefit

This occurs when:

- Welfare benefit is generous
- Individual places high value on leisure

> ⚠ **Warning**
>
> This is the problematic effect of welfare programs

Discussion

Why might someone who was working 20 hours per week before welfare was introduced choose to stop working entirely, while someone working 40 hours per week continues to work the same amount?

Negative Income Tax: Overview

What is NIT?

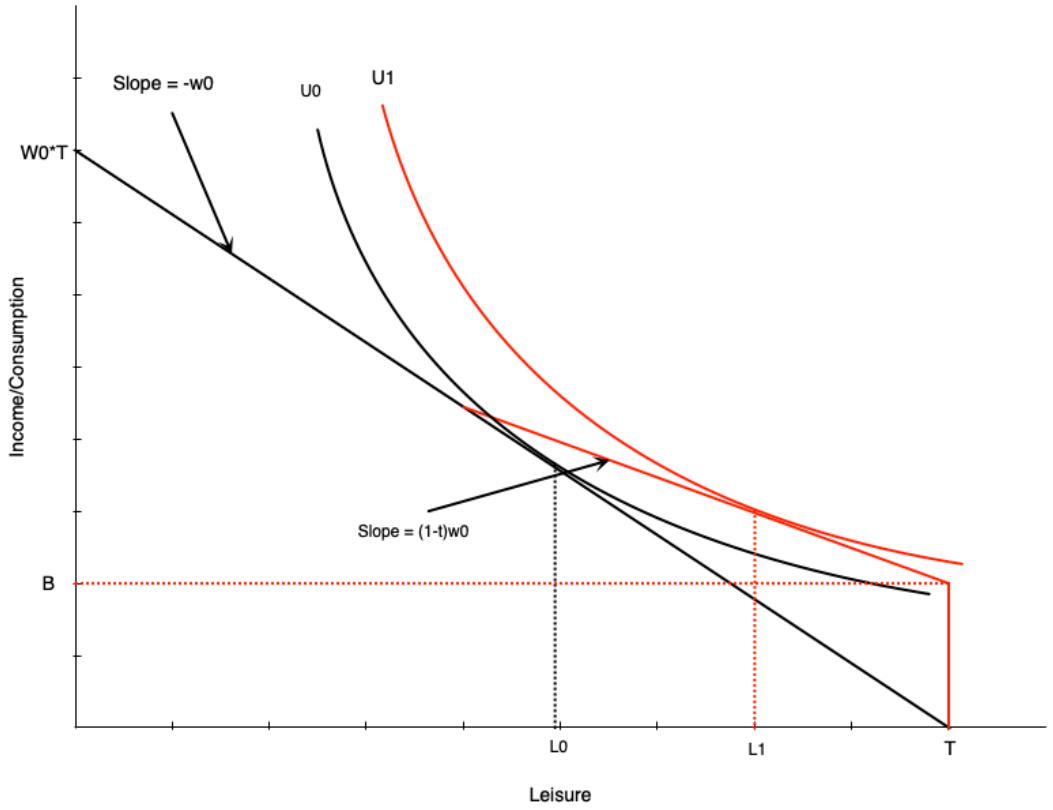
- Welfare program where benefit reduced at rate < 100% of earnings
- Example: guarantee z of \$10,000, reduced by 50¢ per dollar earned

> **Advantages**
>
> Provides income support with fewer work disincentives. Still maintains incentive to work since benefit reduces slowly.

How it relates to basic income:

- A targeted basic income guarantee (BIG) is an NIT: a guarantee plus a clawback
- A universal basic income (UBI) pays everyone the same amount, so it works like a demogrant

Negative Income Tax: Budget Constraint



NIT has two components:

- Guarantee z at zero hours of work
- Implicit tax rate t on earnings, until the benefit is gone at the breakeven point B

Effect on labour supply:

- Pivots budget line up vs. pure welfare
- Work hours reduced from H_0 to H_1
- Person still works some hours

Important

Work disincentive less than pure welfare

Negative Income Tax: Income and Substitution Effects

Income effects:

- Benefit increases non-labour income → increases leisure ↑
- Tax has negative income effect → reduces leisure ↓
- Net: benefit effect dominates → leisure ↑

Substitution effect:

- Tax rate reduces effective wage
- Makes leisure relatively cheaper → reduces work hours ↓

> **Overall Effect**
>
> Negative effect on work hours, but smaller than pure welfare

Negative Income Tax: Math

The budget constraint with NIT:

$$I = \begin{cases} wH + z - twH, & \text{if } z - twH \geq 0 \\ wH, & \text{otherwise} \end{cases}$$

Interpretation:

- Income = earnings (wH) + net benefit $G = z - twH$, as long as $z - twH \geq 0$
- Otherwise just earnings (wH)
- The same thing in one line: $I = z + (1 - t)wH$, with $G = \max(z - twH, 0)$

Breakeven point B: the benefit reaches zero when $wH = \frac{z}{t}$

Basic Income in Canada: MINCOME

Manitoba Basic Annual Income Experiment, 1975–1979

- A targeted basic income, so an NIT, offered to eligible families in two places
  - Winnipeg: families assigned to plans or to a control group
  - Dauphin: a saturation site, where every resident was eligible, with a 50% clawback
- Plans varied both the guarantee and the clawback rate
- The idea had real policy weight: the Macdonald Commission (1985) recommended a national NIT for low-income families

MINCOME: Results

Labour supply effects:

- Hours fell, but not by much (Hum & Simpson 1991)
  - Men: 0.8–1.6%
  - Married women: 2.4–3%
- The US income maintenance experiments found much larger drops for husbands and wives (Hum & Simpson 1993)

The debate continues:

- Basic income is back on the policy agenda in Canada
- Koebel & Pohler (2019): grow the Canada Workers Benefit into a basic income run through the tax system
- Open questions: cost, financing, and federal–provincial cooperation

Other Ways to Reduce Work Disincentives

Policy options:

1. Reduce the guarantee z
   - But then support may be inadequate for those who cannot work
2. Move toward an NIT
   - Clawback below 100%; most provinces now let recipients keep some earnings
3. Change preferences
   - Stigma, or support for getting into work: flatter indifference curves
   - Difficult to achieve
4. Raise the wage
   - Training, job information, minimum wages, unions
   - Or subsidize the wage directly: next section

Wage Subsidies and Refundable Tax Credits

Wage Subsidies: Theory

What is a wage subsidy?

- Payment to workers that increases their wage rate
- Example: $5/hour subsidy increases $15/hour wage to $20/hour

Effect on budget constraint:

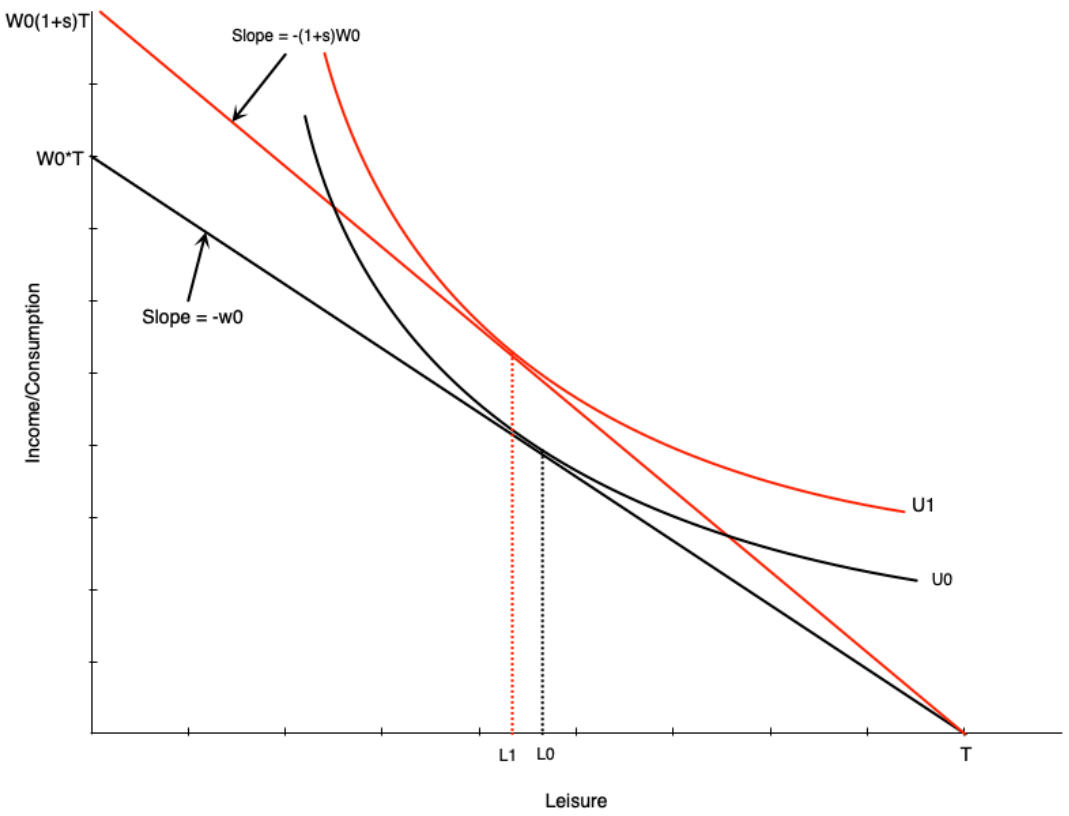
- Budget line pivots upward

Labour supply effects:

- Income effect → increases leisure ↑
- Substitution effect → decreases leisure ↓

> ⚠ **Warning**
>
> Overall effect on work hours is ambiguous

Wage Subsidies: Diagram



Budget line pivots up with subsidy

- Graph shows person working more
- But effect is theoretically ambiguous
- Income and substitution effects work in opposite directions

Wage Subsidies: Policy Implications

Advantages

- Less negative work incentives than welfare or NIT
- No reduction in effective wage rate

Disadvantages

- Do not provide income support to non-workers
- High earners qualify too, so it is a very expensive way to reach those in need

In practice:

- Pure wage subsidies are rare, because of that cost
- The fix is to target: pay the subsidy only to low earners
  - Canada's version is the Canada Workers Benefit

Think About It

A negative income tax reduces the benefit as you earn more, while a wage subsidy increases your pay for each hour worked. Which program would you expect to have a larger positive effect on labour force participation among non-workers, and why?

Refundable Tax Credits

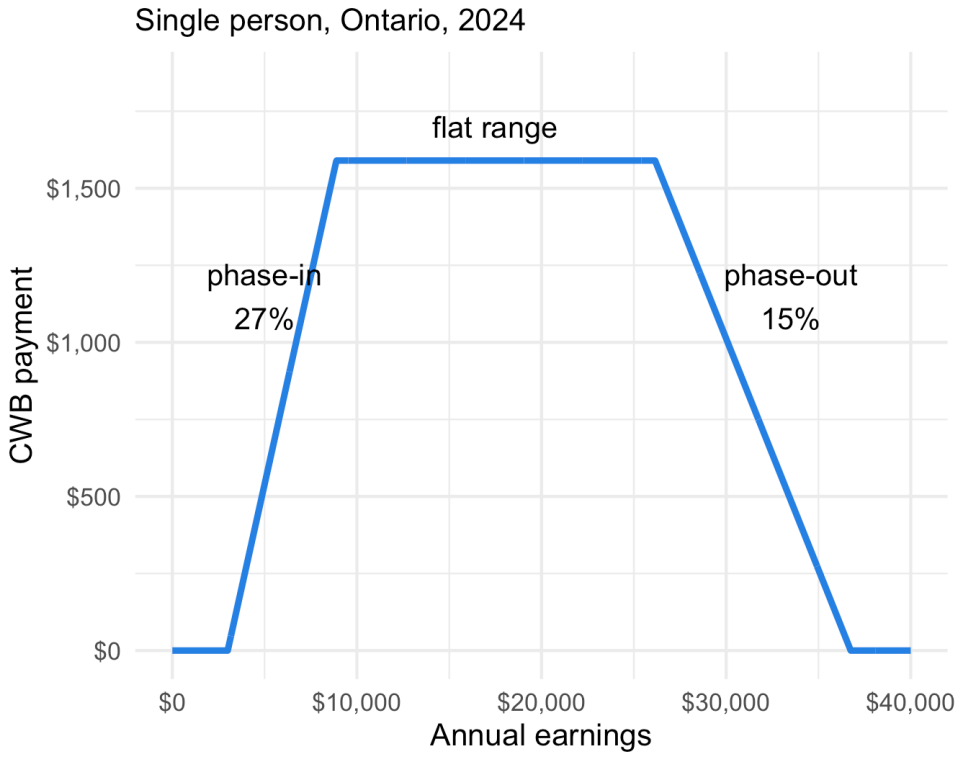
Two types of tax credits:

1. Non-refundable: Can reduce tax owed to zero (but not below)
2. Refundable: Can reduce tax owed below zero (results in payment)

> **Income-Tested Wage Subsidies Paid as Refundable Credits**
>
> - Canada Workers Benefit (CWB) - Canada, since 2007 (first called the WITB)
> - Earned Income Tax Credit (EITC) - US, since 1975, now one of the largest US income supports
> - Similar programs in the UK (Working Families Tax Credit) and New Zealand (In-Work Tax Credit)

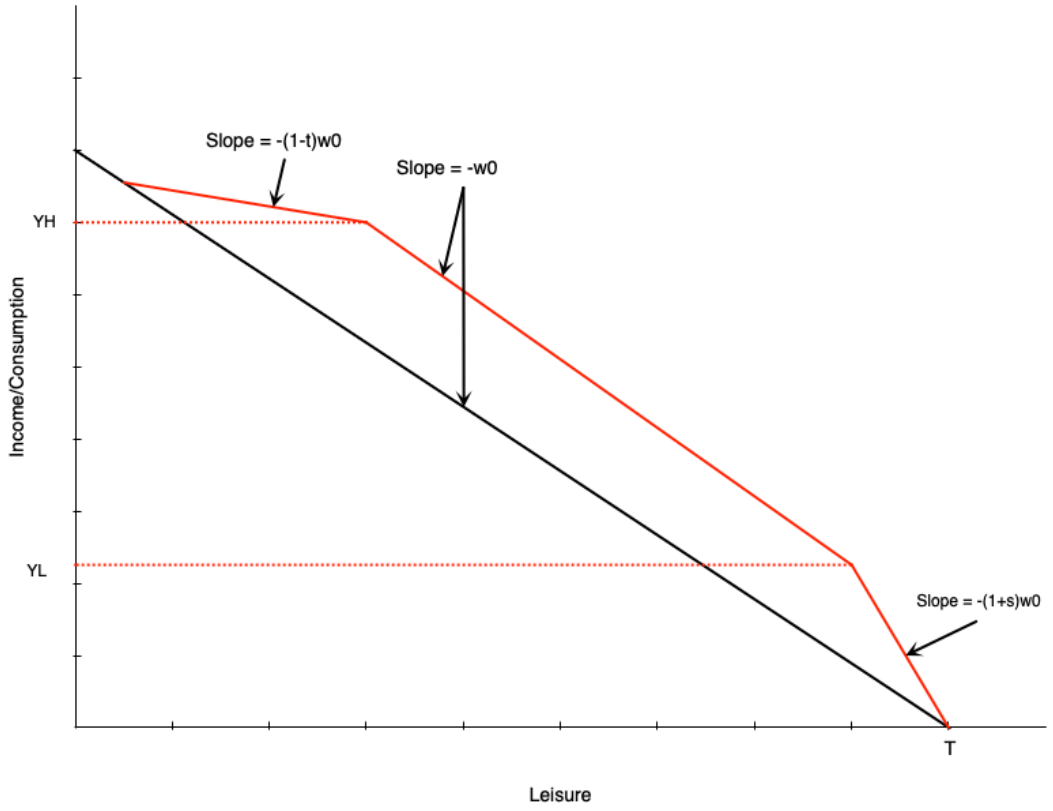
- Popular across the political spectrum: targeted, pro-work, and cheap to run through the tax system, with no caseworkers
- Source: Nichols & Rothstein (2015); Blundell (2006)

The Canada Workers Benefit, 2024



- Nothing until earnings pass $3,000
- Phase-in: 27 cents per dollar earned above $3,000
- Flat range: the $1,590 maximum, reached at about $8,900 of earnings
- Phase-out: clawed back at 15% of earnings above $26,149
- Gone entirely at $36,749
- A small program next to the EITC: low benefits and low thresholds

Tax Credits: Budget Constraint



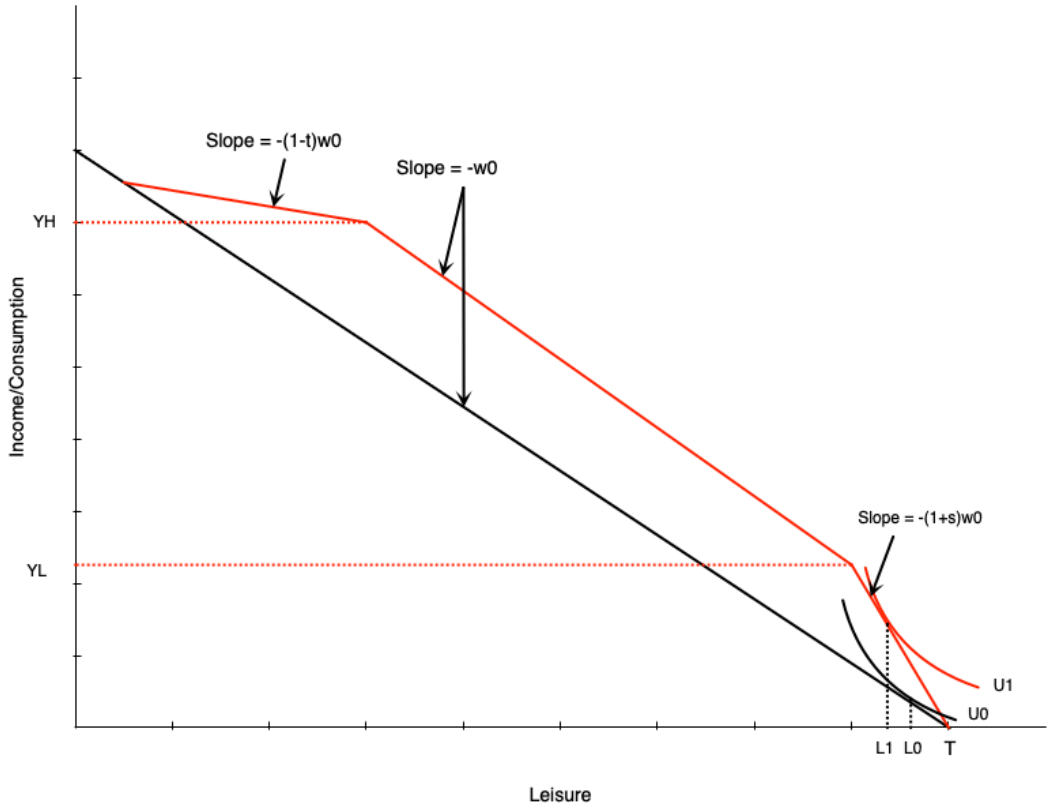
Three segments:

1. Phase-in: Subsidy proportional to earned income, up to income I_L
2. Flat range: Maximum benefit level, between I_L and I_H
3. Phase-out: Benefit reduced with earned income above I_H

Combines elements of:

- Wage subsidy (phase-in)
- Demogrant (flat range)
- NIT (phase-out)

Tax Credits: Effect in Phase-In Range

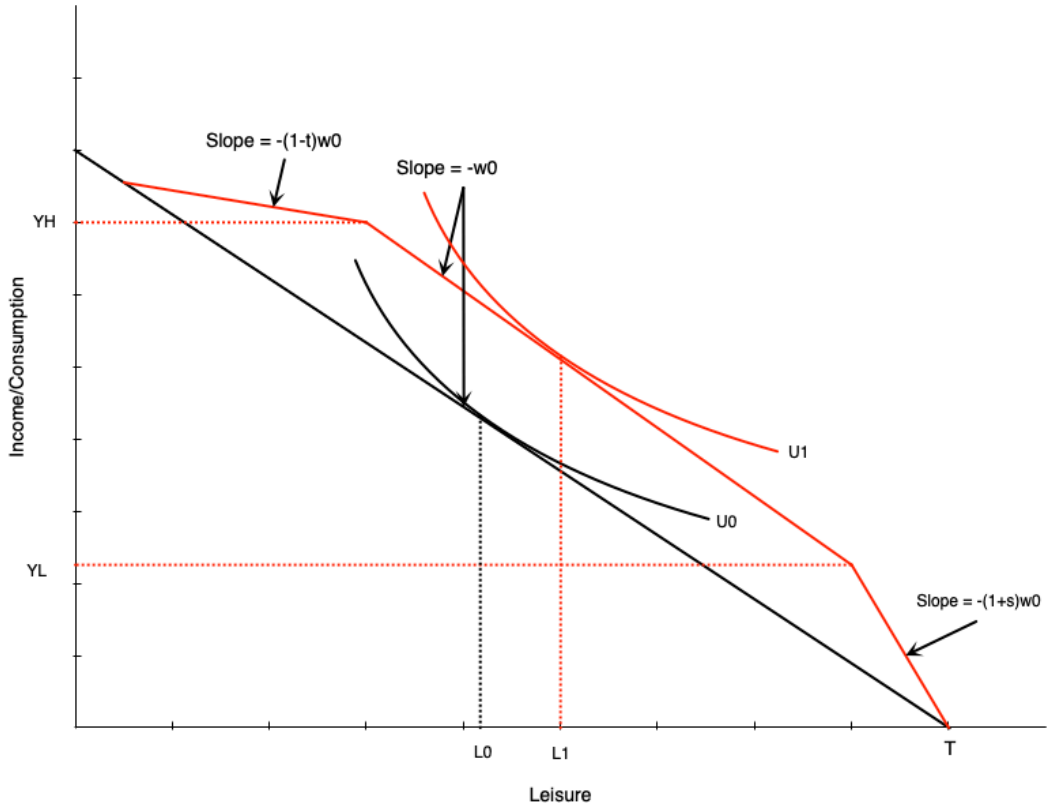


Person in phase-in range:

- Sees increase in effective wage
- Acts like pure wage subsidy
- Income and substitution effects in opposite directions

> **Warning**
>
> Effect on hours is ambiguous

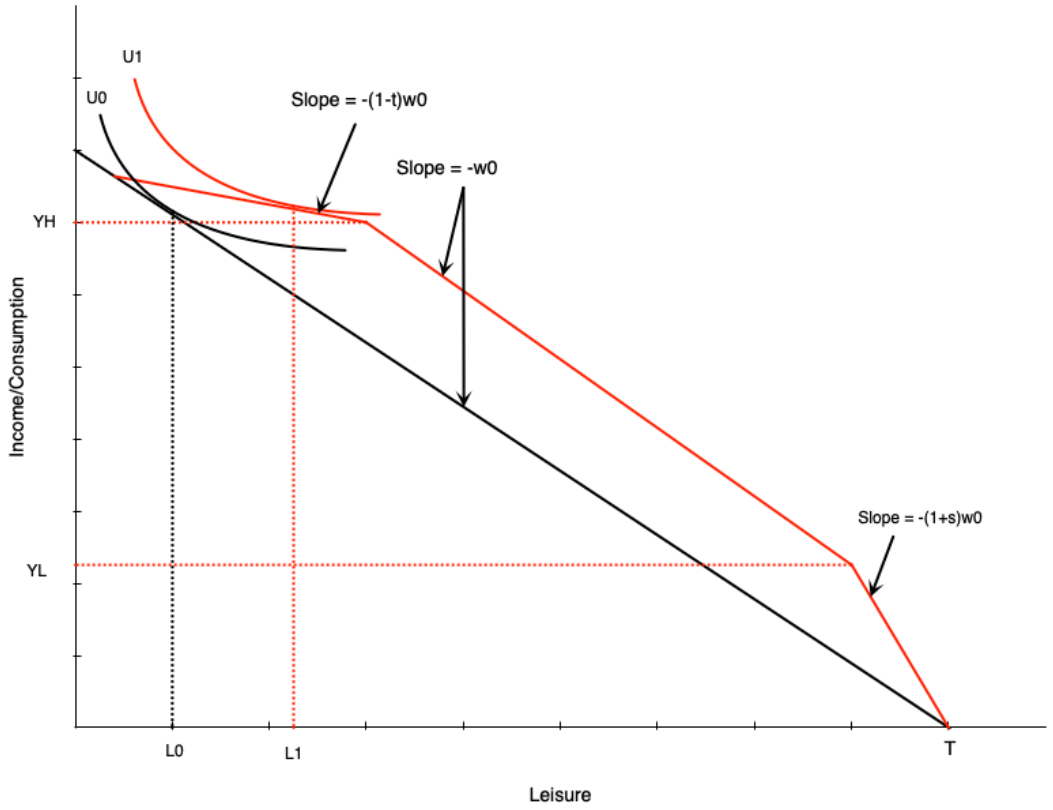
Tax Credits: Effect in Flat Range



Person initially in the flat range:

- Tax credit gives maximum benefit
- Acts like pure income effect
- If leisure is normal good → reduce work hours

Tax Credits: Effect in Phase-Out Range



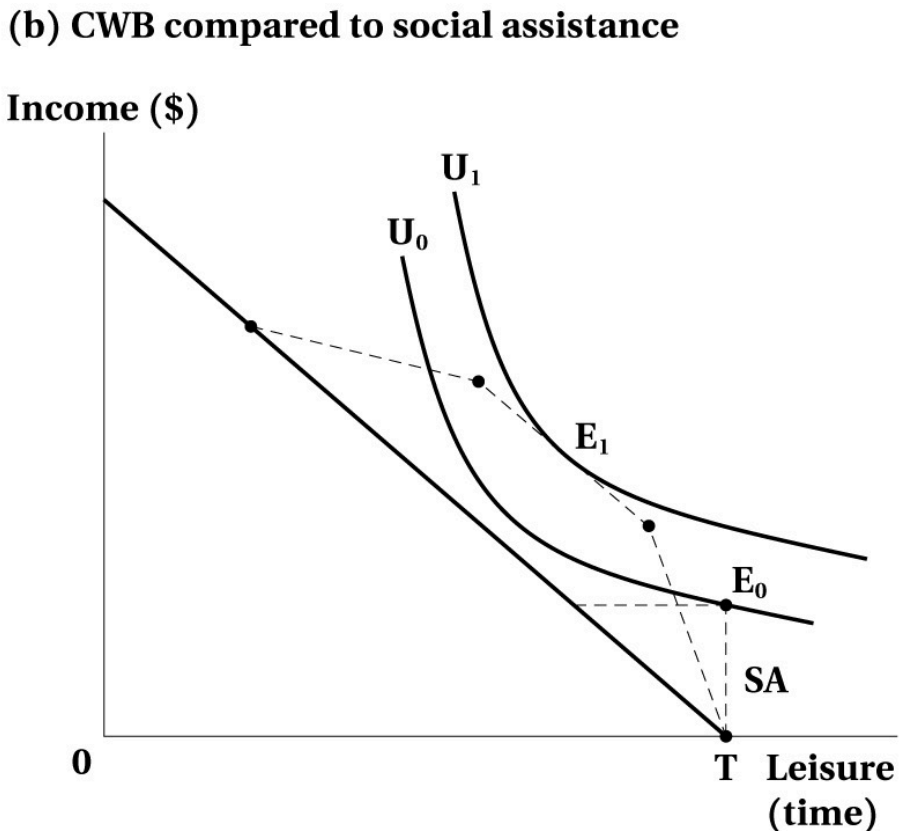
Person in phase-out range:

- Similar to negative income tax
- Income effect → reduce hours ↓
- Substitution effect → reduce hours ↓

Important

Effect on work is unambiguously negative

Tax Credits and Participation



Source: Benjamin et al. (2026), Figure 4.8(b)

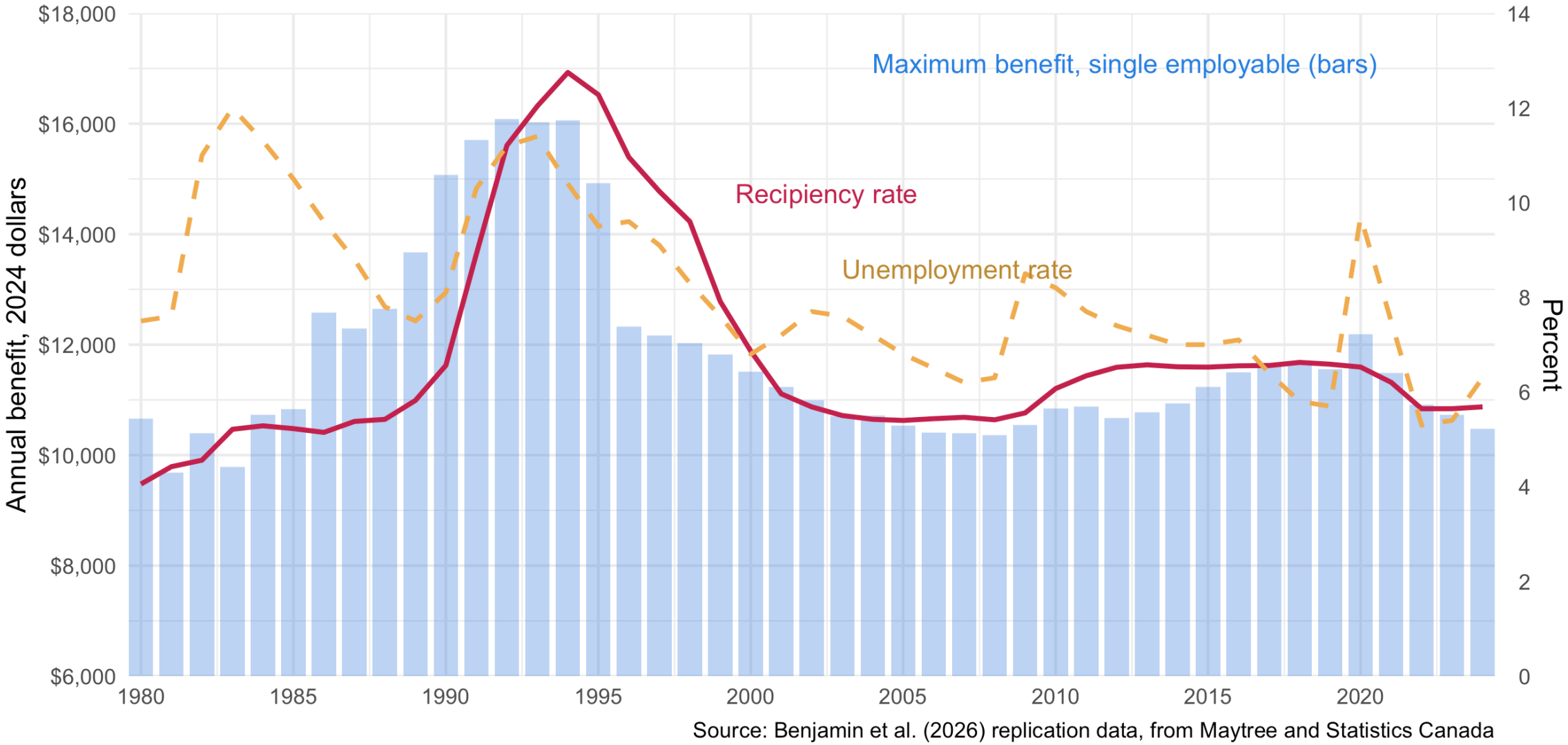
Key Result

Effect on hours is ambiguous, but the incentive to participate rises

- For a non-worker, the phase-in raises the return to the first hours of work
- Against social assistance with a 100% clawback, the contrast is sharper
  - Without the credit: no work, collect assistance (E_0)
  - With it: work, and reach a higher indifference curve (E_1)

Empirical Evidence of Incentive Effects

Social Assistance in Ontario, 1980–2024



Source: Benjamin et al. (2026) replication data, from Maytree and Statistics Canada

Correlation Is Not Causation

- Benefits and recipiency move together, but which way does causation run? We saw this problem in Chapter 2
- Reverse causation: a big caseload is a big constituency, and governments raised benefits to match
- Confounding: unemployment drives both; recessions push people onto assistance, booms pull them off
- Berg & Gabel (2015): across provinces, work requirements did the most to cut caseloads, 1986–2009

Key Definitions

Nonexperimental sources: responses to real policy changes, compared across people, places, or time

Experimental sources: programs that randomly assign who is offered the treatment

Quebec: A Discontinuity at Age 30

- Until August 1989: Quebec paid singles and couples without children far less if they were under 30
  - Age 29: $185 a month; age 30: $507 a month
- Someone a few months short of 30 is otherwise much like someone a few months past it
  - Comparing the two is a regression discontinuity design, as in Chapter 2
- Results:
  - Fortin et al. (2004): the higher benefit led to much longer spells on assistance
  - Lemieux & Milligan (2008): employment of male high-school dropouts fell about 4 percentage points past age 30 (1986)

> **Note**
>
> Two nonexperimental studies, one conclusion: higher benefits reduce work

Child Benefits and Work

- National Child Benefit, 1998: some provinces effectively cut the clawback for welfare recipients
  - Work incentives improved, explaining much of the fall in caseloads from 1997 to 2000
- UCCB, 2006: compare parents of children 5 and under to parents of older children, before and after
  - A difference-in-differences design, as in Chapter 2
  - Married mothers cut both participation and hours, as the demogrant model predicts
  - From the paper itself: participation of lower-educated married mothers fell 3.2 percentage points
  - Divorced mothers worked *more*
- UCCB expansion and the CCB: large cuts in poverty, no sign of lower labour supply
- Source: Milligan & Stabile (2007); Schirle (2015); Koebel & Schirle (2016); Baker et al. (2023)

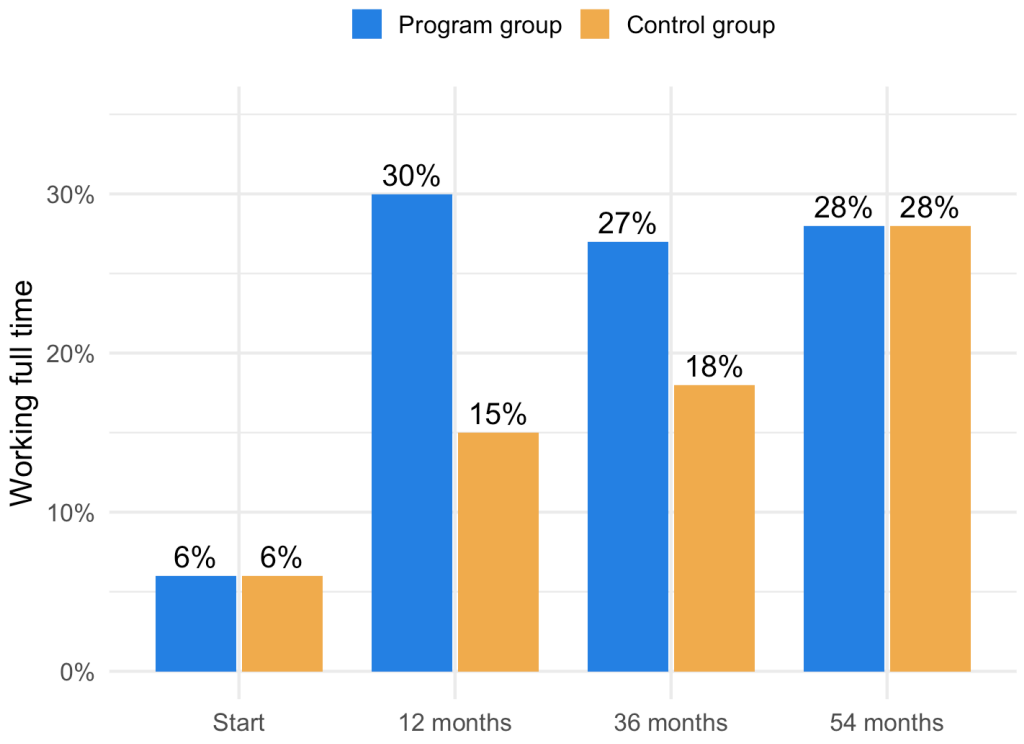
The Self-Sufficiency Project

Key Definition

Self-Sufficiency Project (SSP): a randomized Canadian experiment that paid an earnings supplement to long-term assistance recipients who left for full-time work

- Who: about 5,600 single parents, mostly mothers, on assistance for at least a year
- Where and when: New Brunswick and British Columbia, 1992–1999
- Design: random assignment to a program group or a control group
- The offer: leave assistance for a job of 30+ hours a week within a year, and collect a supplement
  - Half the gap between earnings and a benchmark of $30,000 (NB) or $37,000 (BC)
  - Paid for up to three years, as long as the work stays full time
  - Roughly doubled what assistance paid

SSP: Results



- Naive estimate: full-time work in the program group went from 6% to 30%, so 24 points
- But the control group got to 15% with no offer at all
- Experimental estimate: 30 − 15 = 15 points
- The gap closed after the offer ended
  - 9 points at 36 months
  - Gone by 54 months; assistance receipt equal by month 69

SSP: Why the Effect Faded

- Card & Hyslop (2005): the time limits created two incentives
  - Establishment incentive: find full-time work within 12 months to qualify
  - Entitlement incentive: once qualified, keep working to keep collecting
  - Both end when the program ends
- Riddell & Riddell (2014): the program group went back to school less
  - Less high-school completion, college or trade school, and university
  - Work-first policies can trade long-run earning power for short-run jobs

> **Note**
>
> The SSP sped up the move from assistance to work while it ran; it did not change where people ended up

- Source: Card & Hyslop (2005); Riddell & Riddell (2014)

Summary

Summary

Income support programs:

- Provide crucial income support to various groups
- Can create work disincentives in some cases
- Must balance support needs against incentive effects

We examined several programs through the labour supply model:

- Three steps: constraint before, constraint after, compare choices
- Demogrants, social assistance, NIT, wage subsidies, and tax credits like the CWB

Evidence:

- Experiments and natural experiments beat simple correlations
- Both benefit generosity and labour market opportunities predict who relies on assistance

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